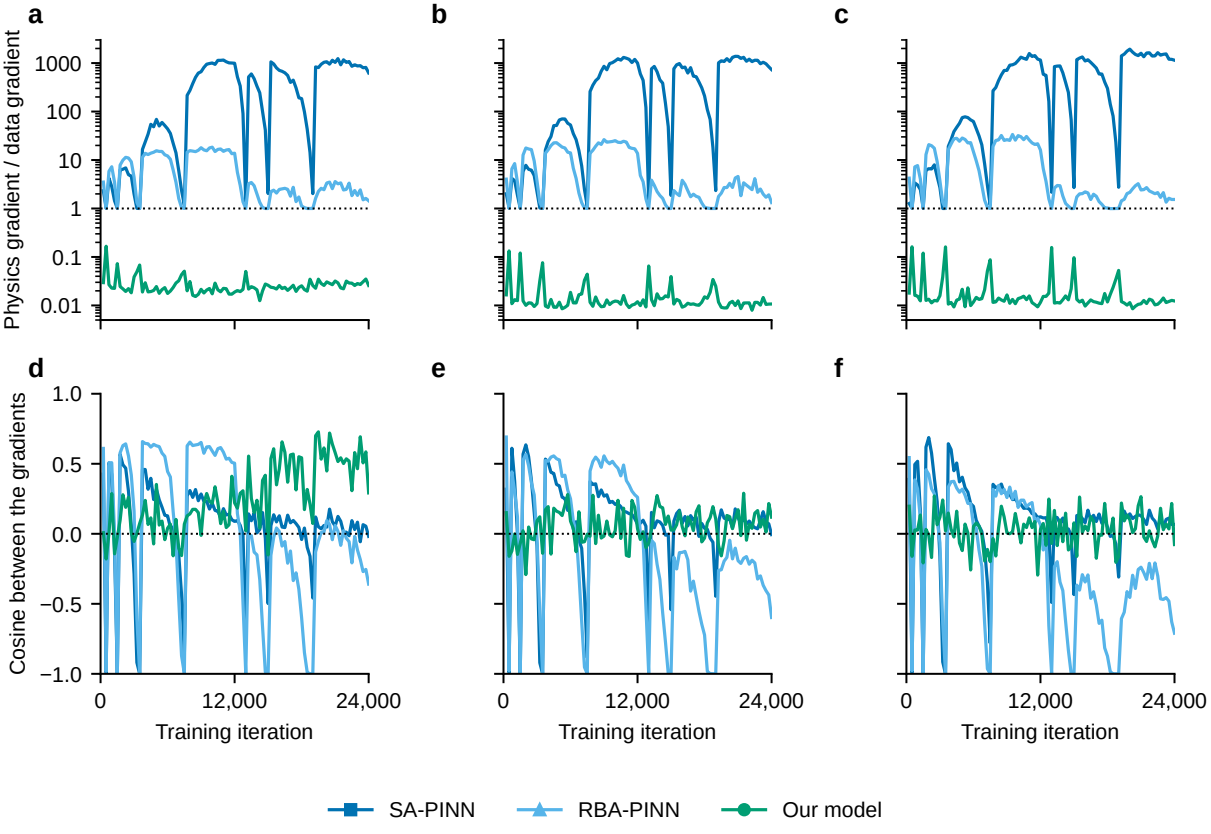
a
b
c
d
e
f
Physics gradient / data gradient
1000
100
10
1
0.1
0.01
Cosine between the gradients
1.0
0.5
0.0
−0.5
−1.0
0
12,000
24,000
Training iteration
SA-PINN
RBA-PINN
Our model